

Unit 2—Lesson 6:

Loops

Loops

for
while

for loops

```
for index in 1...5 {  
    print("This is number \ (index)")  
}
```


for loops

```
for _ in 1...5 {  
    print("Hello!")  
}
```

for loops

```
let names = ["Joseph", "Cathy", "Winston"]  
for name in names {  
    print("Hello \(name)")  
}
```

Hello Joseph
Hello Cathy
Hello Winston

```
for letter in "ABCDEFG" {  
    print("The letter is \(letter)")  
}
```

The letter is A
The letter is B
The letter is C
The letter is D
The letter is E
The letter is F
The letter is G

for loops

```
for (index, letter) in "ABCDEFGG".enumerated() {  
    print("\(index): \(letter)")  
}
```

0: A
1: B
2: C
3: D
4: E
5: F
6: G

- `enumerated()` gives you the item and the index

NOTE: `(index, letter)` is a tuple.

for loops

Iterating Over a dictionary

```
let vehicles = ["unicycle" : 1, "bicycle" : 2, "tricycle" : 3, "quad bike" : 4]
for (vehicleName, wheelCount) in vehicles {
    print("A \(vehicleName) has \(\wheelCount) wheels")
}
```

A tricycle has 3 wheels
A bicycle has 2 wheels
A quad bike has 4 wheels
A unicycle has 1 wheels

while loops

```
var numberOfLives = 3

while numberOfLives > 0 {
  playMove()
  updateLivesCount()
}
```


while loops

```
var numberOfLives = 3

while numberOfLives > 0 {
    print("I still have \(numberOfLives) lives.")
}
```

while loops

```
var numberOfLives = 3
var stillAlive = true

while stillAlive {
    print("I still have \(numberOfLives) lives.")
    numberOfLives -= 1
    if numberOfLives == 0 {
        stillAlive = false
    }
}
```

Control transfer statements

```
for counter in -10..10 {  
    print(counter)  
    if counter == 0 {  
        break  
    }  
}
```

```
-10  
-9  
...  
0
```


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Lab: Loops



Open and complete the exercises in Lab - Loops.playground.

